Bioinfo Assignment 2

Bohan YANG
Student ID: 2330016056

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Question 1

Listing 1: q1.py

```python
import networkx as nx
import numpy as np


def main():
    G = nx.barabasi_albert_graph(n=30, m=2, seed=42)

    A = nx.to_numpy_array(G, dtype=int, nodelist=sorted(G.nodes()))
    np.savetxt("adjacency_matrix.csv", A, fmt="%d", delimiter=",")

    nx.write_graphml(G, "network.graphml")


if __name__ == "__main__":
    main()
```

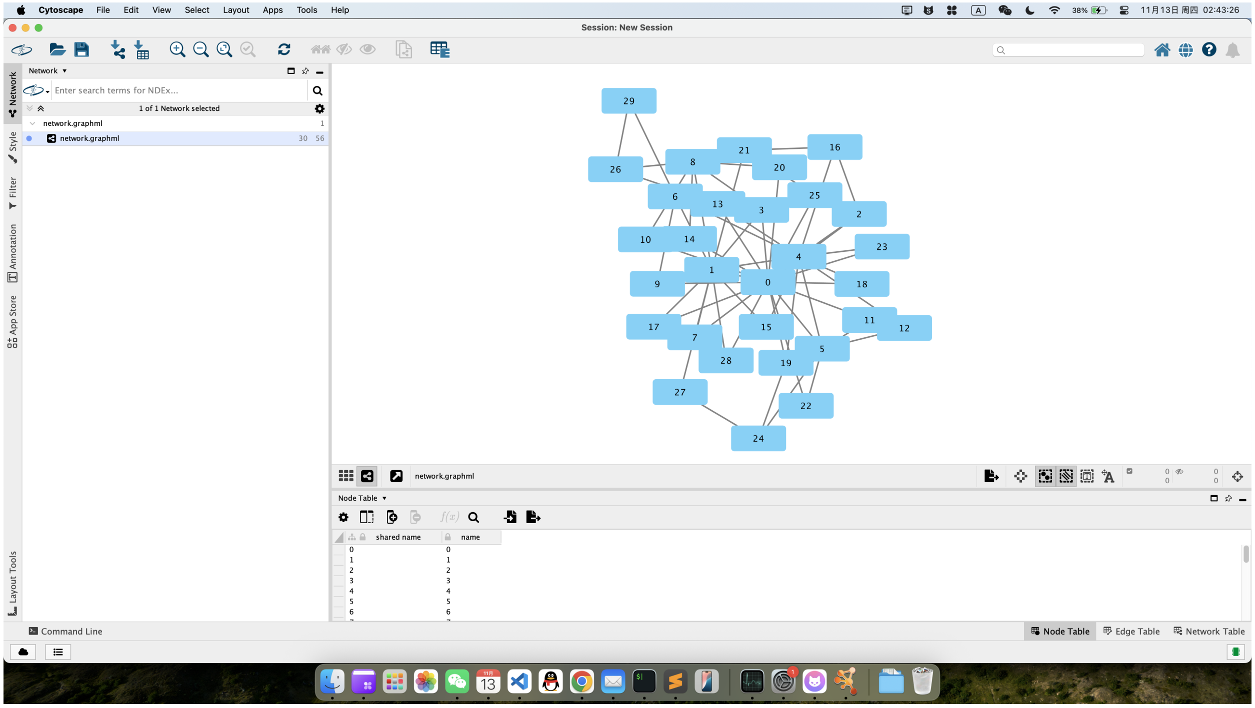

Figure 1: Cytoscape Barabási–Albert network (n=30)

Listing 2: Adjacency Matrix (CSV format)

```
0,1,1,1,0,1,0,1,0,1,1,1,0,1,1,0,0,1,1,1,1,0,1,1,0,1,0,0,1,0
1,0,0,1,1,0,1,1,1,0,0,0,0,0,0,1,0,1,0,0,0,1,0,0,0,0,0,1,1,0
1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0
1,1,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,1,1,0,0,1,1,0,0,0,0,0,1,1,0,1,1,0,1,1,0,0,0,1,0,0,0,0,0,0
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1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0
1,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
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1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
```

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0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,1,0,0
1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1
0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0
1,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0
```

Question 2

1. Degree of x:

$$\deg(x) = 5$$

since x is connected to nodes $\{1, 2, 3, 4, 5\}$.

2. Betweenness centrality of x:

There are $\binom{5}{2} = 10$ possible pairs of leaf nodes.

The number of shortest paths passing through x is 9.

$$B(x) = 9$$

Normalized betweenness:

$$B_{\text{norm}}(x) = \frac{9}{10} = 0.9$$

3. Characteristic path length of the network:

Total number of node pairs:

$$\binom{6}{2} = 15$$

Shortest path lengths:

$$\text{distance } = 1 \text{ for 6 pairs: } (x, 1), (x, 2), (x, 3), (x, 4), (x, 5), (4, 5)$$

$$\text{distance } = 2 \text{ for 9 pairs: all other leaf–leaf pairs.}$$

Hence,

$$L = \frac{6 \times 1 + 9 \times 2}{15} = \frac{24}{15} = 1.6$$

$$\boxed{\deg(x) = 5, \quad B(x) = 9 \ (B_{\text{norm}} = 0.9), \quad L = 1.6}$$

Question 3

Listing 3: q3.py

```
import numpy as np
import pandas as pd

genes = ["G1", "G2", "G3", "G4"]
MI = pd.DataFrame([
    [1.24, 1.01, 0.69, 0.69],
    [1.01, 1.56, 1.01, 0.78],
    [0.69, 1.01, 1.01, 0.46],
    [0.69, 0.78, 0.46, 1.01]
], index=genes, columns=genes)

# calc Zi
Zi = MI.apply(lambda x: (x - x.mean()) / x.std(ddof=0), axis=1)
Zi = Zi.clip(lower=0)

# calc Zj
Zj = MI.apply(lambda x: (x - x.mean()) / x.std(ddof=0), axis=0)
Zj = Zj.clip(lower=0)

# calc CLR mat
CLR = np.sqrt(Zi ** 2 + Zj ** 2)
CLR = pd.DataFrame(CLR, index=genes, columns=genes)

pairs = []
for i in range(len(genes)):
    for j in range(i + 1, len(genes)):
        pairs.append((genes[i], genes[j], CLR.iloc[i, j]))

top3 = sorted(pairs, key=lambda x: x[2], reverse=True)[:3]

print("Zi mat\n", Zi.round(3))
print("\nZj mat\n", Zj.round(3))
print("\nCLR mat\n", CLR.round(3))
print("\nTop 3 GGI")
for g1, g2, val in top3:
    print(f"{g1}-{g2}: {val:.3f}")
```

Row-wise positive z-scores (Z_i):

$$Z_i = \begin{bmatrix} 1.432 & 0.441 & 0 & 0 \\ 0 & 1.637 & 0 & 0 \\ 0 & 0.937 & 0.937 & 0 \\ 0 & 0.228 & 0 & 1.396 \end{bmatrix}.$$

Column-wise positive z-scores (Z_j):

$$Z_j = \begin{bmatrix} 1.432 & 0 & 0 & 0 \\ 0.441 & 1.637 & 0.937 & 0.228 \\ 0 & 0 & 0.937 & 0 \\ 0 & 0 & 0 & 1.396 \end{bmatrix}.$$

CLR matrix:

$$CLR = \begin{bmatrix} 2.025 & 0.441 & 0 & 0 \\ 0.441 & 2.315 & 0.937 & 0.228 \\ 0 & 0.937 & 1.325 & 0 \\ 0 & 0.228 & 0 & 1.974 \end{bmatrix}.$$

Top 3 GGI

$$\begin{aligned} &\textbf{1.}\ (G2, G3) &:\ & CLR_{23} = 0.937, \\ &\textbf{2.}\ (G1, G2) &:\ & CLR_{12} = 0.441, \\ &\textbf{3.}\ (G2, G4) &:\ & CLR_{24} = 0.228. \end{aligned}$$

Question 4

(a) What is the major mutation signature contribute to AML?

According to the given matrix, AML is associated with **Signature 1** and **Signature 5**.

$\boxed{\textbf{Signature 1}}$ is the most dominant across all cancer types and typically shows stronger contribution in AML.

(b) What is the major mutation type related to AML?

$\boxed{\textbf{C>T}}$ is accounted for more than 10% of all.

(c) Signatures Dominated by T>A

$\boxed{\textbf{Signature 22 and Signature 27}}$ are dominated by T>A, which is more than 10%.

Associated Cancer Types: Liver cancer, Urothelial carcinoma (Signature 22); Kidney clear cell carcinoma (Signature 27).

(d) Model Design

Feature Representation

Each patient can be represented by:

- **Raw mutation spectrum** (96-dimensional trinucleotide mutation frequencies), or
- **Signature contributions** obtained from NMF (H vector).

Model Choice

As far as the task is concerned, we can use a two-step model:

$$\textbf{NMF decomposition} \Rightarrow \textbf{Random Forest classifier}$$

Procedure

1. Build matrix X from patient mutation profiles.
2. Apply NMF to obtain H (signature weights per patient).
3. Split the data into training and test sets.
4. Train a Random Forest or Logistic Regression model using H as input and cancer type as target.
5. For a new patient, compute mutation profile $\rightarrow$ project to H space $\rightarrow$ predict cancer type.

Why This Model Fits the Task

- NMF provides biologically interpretable latent features (mutation signatures).
- The classifier captures statistical relationships between signature patterns and cancer types.
- The model can generalize to unseen patients with similar mutational patterns.
- Probability outputs allow risk estimation and uncertainty quantification.